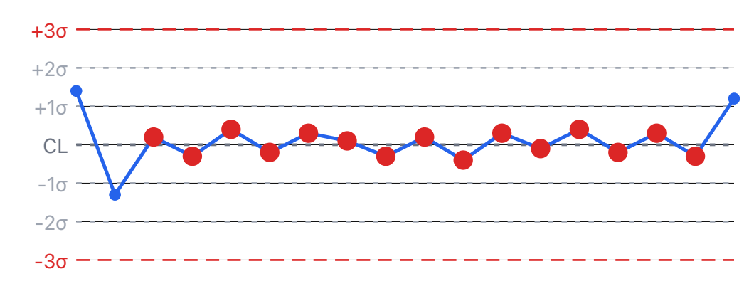
AeroPrint QC

SPC and capability screening for 3D-printed titanium brackets

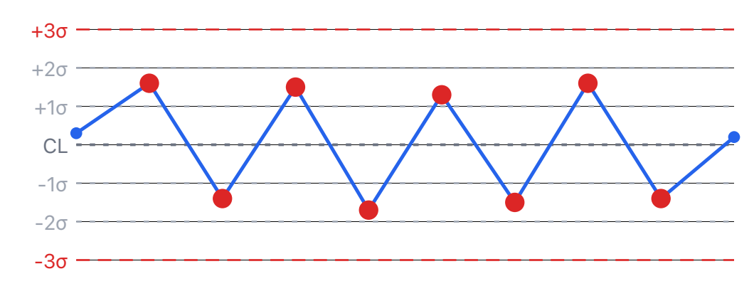
An aerospace contractor 3D-printing titanium brackets is failing client strength tests, and does not track enough process parameters against final part strength. This tool monitors and flags *when* the process goes out of control and *whether* it can meet the strength requirement, and points to what to investigate.

- Upload prints in production order, or try a built-in demo.
- See **when** the process shifted (X̄, S, I and MR charts, eight special-cause tests).
- See **whether** it is capable (Cpk, Ppk, Cp, Pp, histogram).
- Check every parameter against its tolerance, and find which one moves with strength.
- Get corrective investigations and download a per-print results file.



Fifteen prints in a row all within 1σ of the centre line: suspiciously little variation (for example, mixed data or rounded readings).

Test 8: 8 in a row beyond 1 sigma



Eight prints in a row, all beyond 1σ (either side): the process is jumping between two levels.

Ticking more tests catches smaller shifts but also raises false alarms (a flag when nothing is wrong). On 60 prints of pure noise: test 1 alone about 22%, tests 1+2 about 30%, tests 1,2,5,6 about 45%, all eight about 60%.

5. ADVANCED (OPTIONAL)

The defaults are fine for most uses.

Baseline prints: how many early prints define normal (0 = automatic)

auto

X-bar subgroup size: prints grouped per point on the X-bar and R charts

5

1 · Detect
Watch the charts and the red banner. They tell you *when* the process went out of control.

2 · Diagnose
Check the parameter table and corrective list. They point to *where* to look.

3 · Verify
Fix that variable, upload the new prints, and confirm the charts recover.

SCREENING PASSED: no control signals and Cpk and Ppk both meet the target. Passing is not qualification.

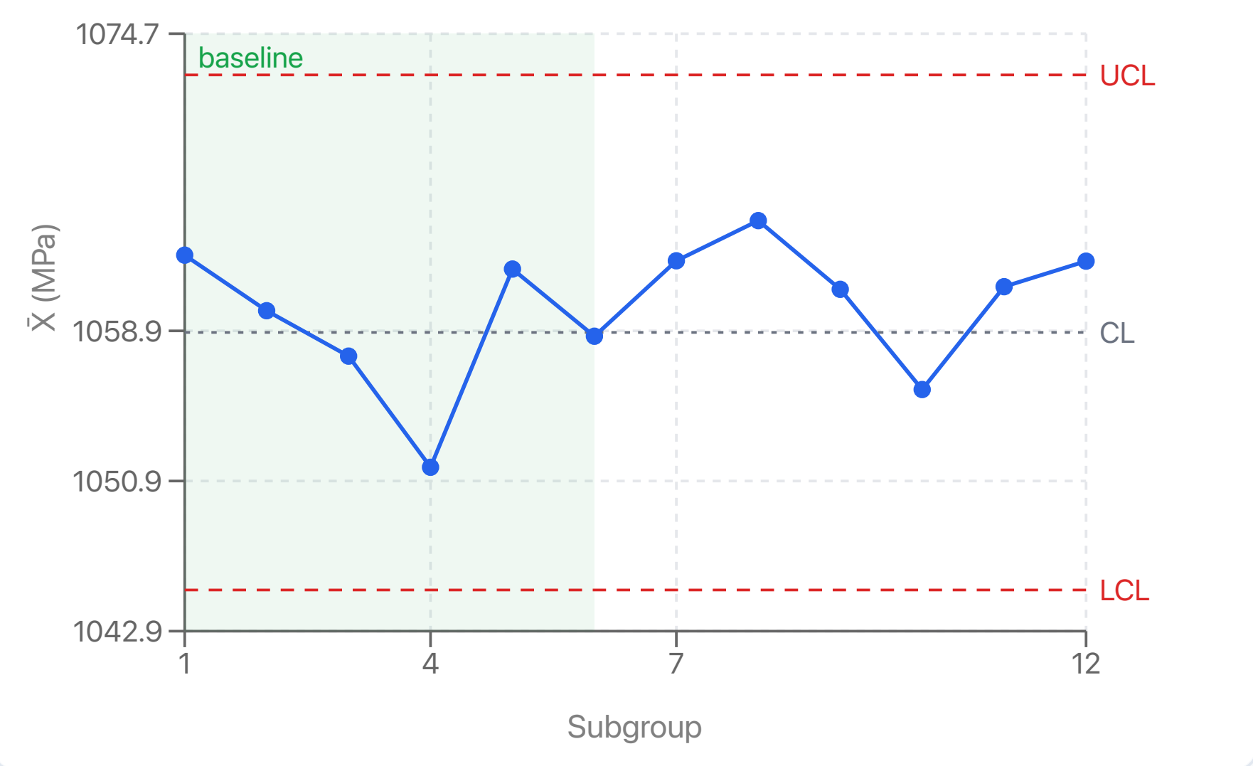
Mean strength	Cpk (short-term)	Ppk (long-term)	Prints below requirement	Out-of-control prints
1060.1 MPa	**2.36**	**2.36**	**0 / 60**	**0**
σ̂ (overall) 8.5 MPa, n = 60	95% CI 1.93 to 2.79	target 1.33	minimum 1000 MPa	none flagged

Both use the same formula, (mean − minimum) ÷ (3 × σ̂). **Cpk** takes σ̂ from print-to-print differences (what the process can do when steady). **Ppk** takes σ̂ from all the data (what it actually did). A Ppk much lower than Cpk means the process drifted along the way.

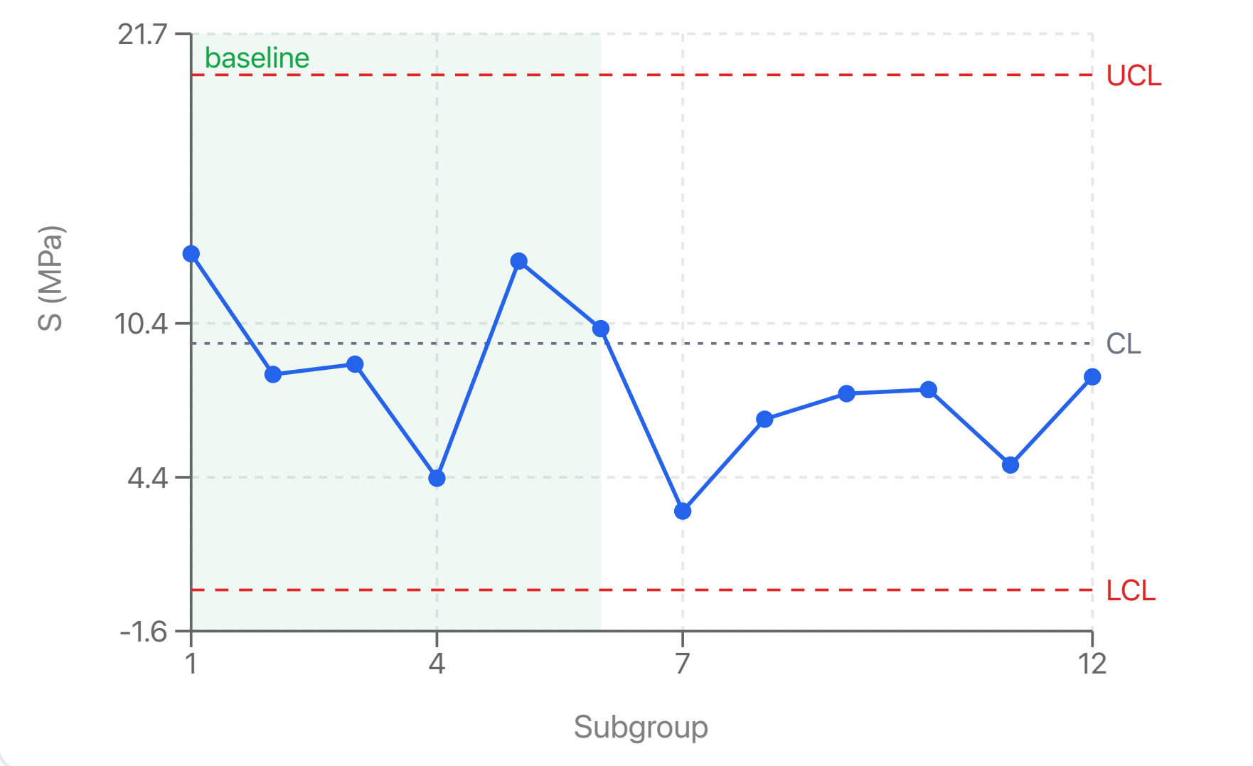
Control charts

We treat the first 30 prints as the process at its normal (shaded green), draw the limits from them, then check whether later points stay inside. The X̄ chart watches the average; the S chart watches the spread.

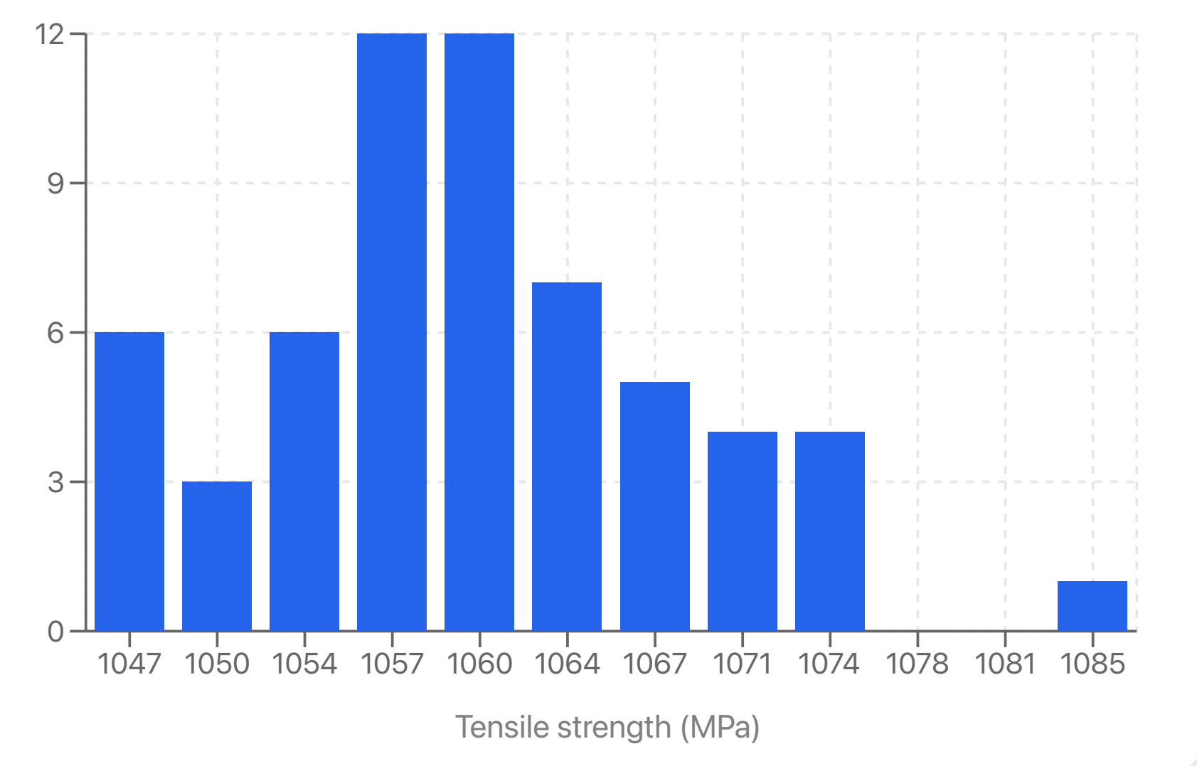
X̄ chart (average of each 5 prints) 12 subgroups



S chart (spread of each 5 prints) standard deviation



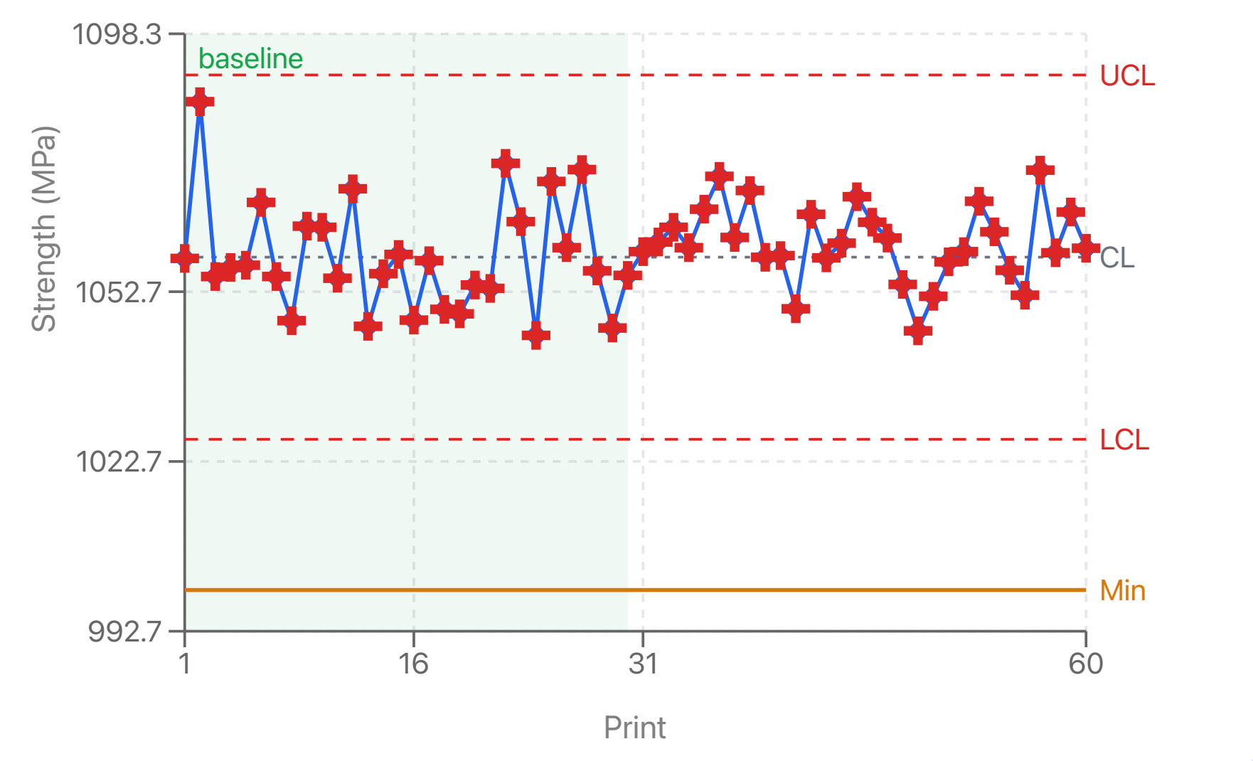
Strength distribution bars below the minimum are red



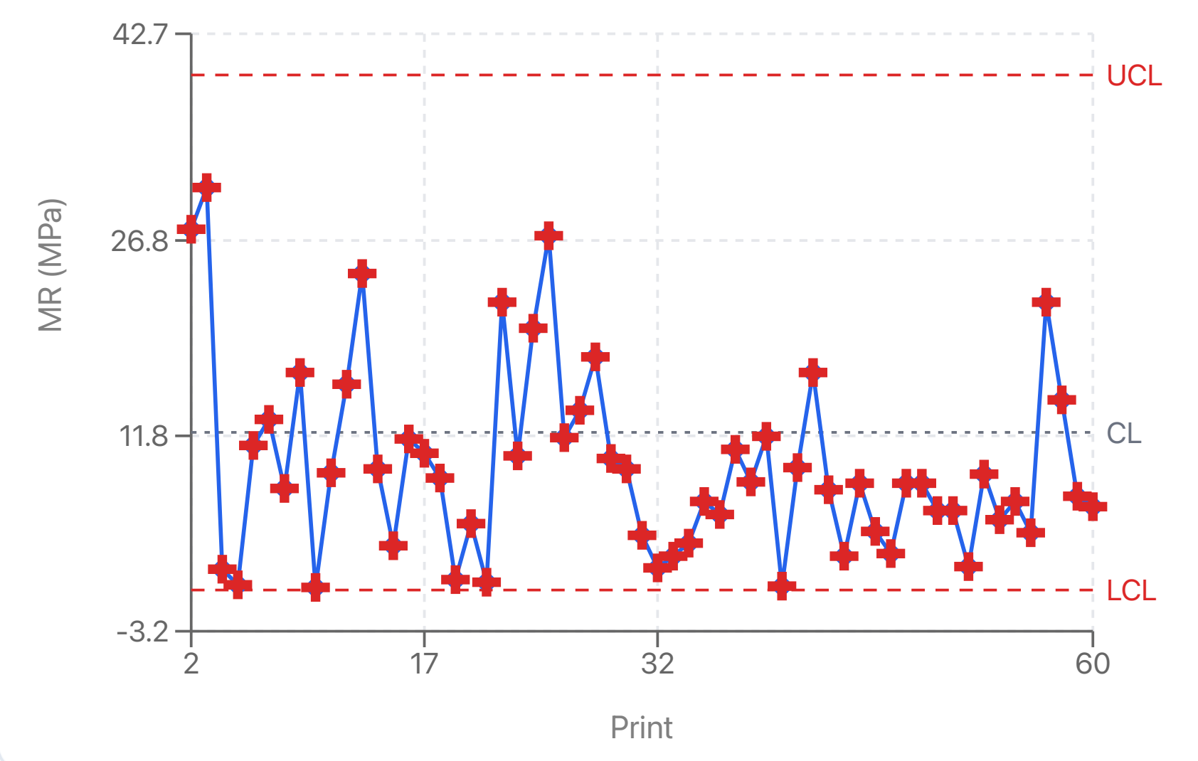
Print-by-print charts (I and MR)

X̄ and S group prints, so a single bad print can hide inside its group. These two plot every print on its own: the I chart shows each print's strength, and the MR chart shows how much each print differs from the one before (moving range). The special-cause tests in the sidebar run on the I chart, and the same moving ranges give σ̂ for Cpk (σ̂ = average MR ÷ 1.128) [2].

I chart (each print) individual strength



MR chart (change from previous print) moving range



Process-parameter inspection

Each print checked against the tolerances in the sidebar, plus capability for each parameter. Cp and Pp need both a lower and an upper limit. Status is judged on Ppk. n/a means it cannot be computed: Cp and Pp need two limits (tensile strength has only a minimum), and a parameter that never varies has no spread.

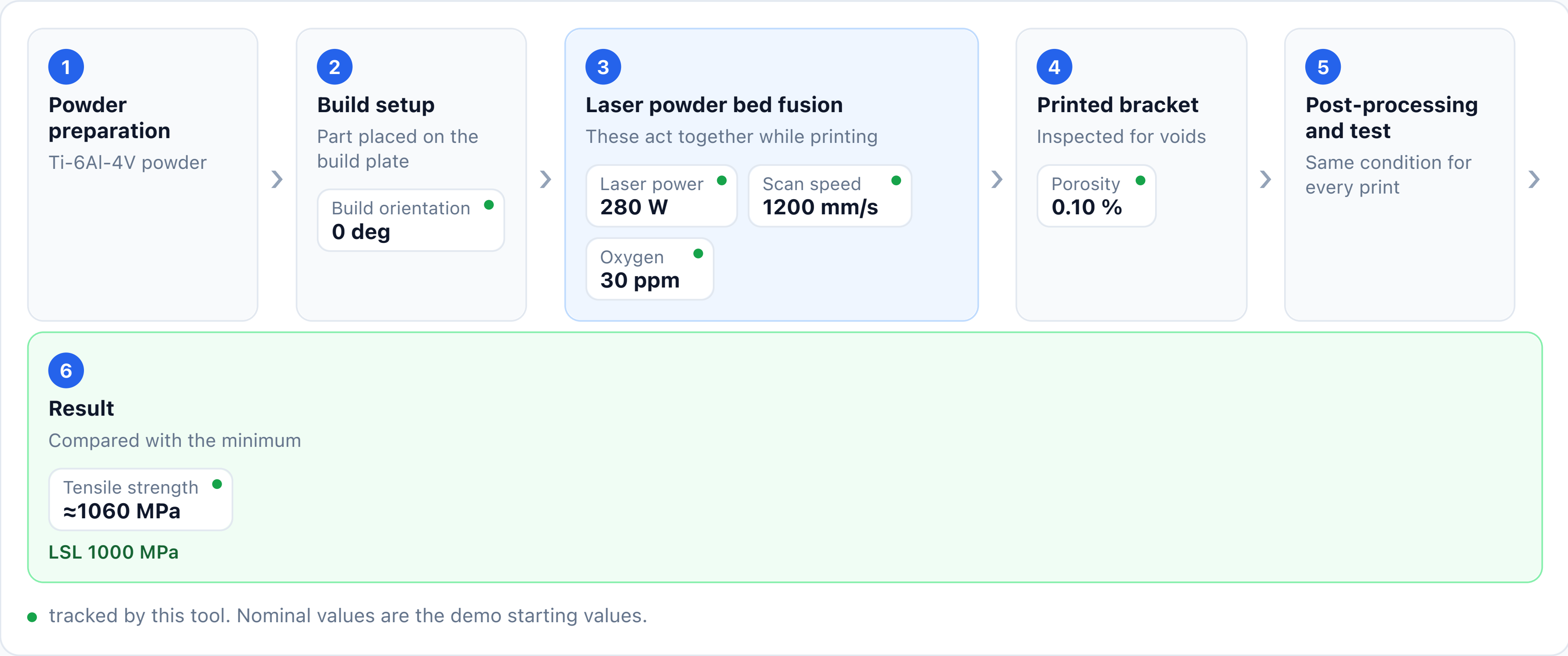
Parameter	Lower	Upper	Mean	% outside	Cp	Cpk	Pp	Ppk	First signal	Status
Tensile strength (MPa)	1000	no limit	1060.095	0.0	n/a	2.36	n/a	2.36	-	ok
Laser power (W)	266	294	279.782	0.0	1.84	1.81	1.65	1.63	-	ok
Scan speed (mm/s)	1140	1260	1199.318	0.0	1.46	1.45	1.50	1.48	-	ok
Oxygen (ppm)	0	50	30.978	0.0	2.49	1.90	2.38	1.81	-	ok
Porosity (%)	0	0.2	0.099	0.0	2.23	2.21	2.11	2.08	-	ok
Build orientation (deg)	0	0	0.000	0.0	n/a	n/a	n/a	n/a	-	ok (constant)

Which parameters move with strength?

Pearson correlation across prints. A lead for investigation, not proof of cause.

Parameter	Pearson r	p-value (exploratory)
Oxygen (ppm)	-0.365	0.004
Porosity (%)	-0.115	0.383
Laser power (W)	0.115	0.383
Scan speed (mm/s)	0.018	0.893

Process flow (reference)



1 **Powder preparation**
Ti-6Al-4V powder

2 **Build setup**
Part placed on the build plate
Build orientation ● **0 deg**

3 **Laser powder bed fusion**
These act together while printing
Laser power ● **280 W**
Scan speed ● **1200 mm/s**
Oxygen ● **30 ppm**

4 **Printed bracket**
Inspected for voids
Porosity ● **0.10 %**

5 **Post-processing and test**
Same condition for every print

6 **Result**
Compared with the minimum
Tensile strength ● **≈1060 MPa**
LSL 1000 MPa

● tracked by this tool. Nominal values are the demo starting values.

Suggested corrective investigations

- Always: confirm the tensile test method (ASTM E8/E8M; ASTM F2924 is the specification for additively manufactured Ti-6Al-4V) and the approved specification, material/process qualification and engineering sign-off. Correlation is not causation.

Inspect and export

Per-print results with pass/fail against every tolerance and the SPC flags.

Download results CSV

#	Strength	Flag	Tests	Laser power (W)	Scan speed (mm/s)	Oxygen (ppm)	Porosity (%)	Build orientation (deg)
1	1058.6	OK		286.1	1169.3	31.5	0.09	0
2	1086.3	OK		279.4	1175.8	29.2	0.084	0
3	1055.4	OK		280.7	1195.8	29	0.088	0
4	1057	OK		278.8	1205.8	29.2	0.117	0
5	1057.4	OK		280.1	1218.5	31.9	0.091	0
6	1068.5	OK		281.6	1223.2	29.1	0.096	0
7	1055.4	OK		277.3	1196.5	33.1	0.11	0
8	1047.6	OK		282	1166.1	33.6	0.083	0
9	1064.3	OK		280.8	1208.4	28.4	0.081	0
10	1064.1	OK		279.8	1216.9	32.6	0.103	0
11	1055.1	OK		279.4	1188.9	32	0.11	0
12	1070.9	OK		277.7	1202.7	21.3	0.112	0
13	1046.6	OK		275.1	1200.7	26.6	0.114	0
14	1055.9	OK		277.3	1208.5	34	0.061	0
15	1059.3	OK		281	1192.7	35.6	0.079	0
16	1047.7	OK		276.9	1216.9	29.9	0.093	0
17	1058.2	OK		285	1209	32.6	0.12	0
18	1049.6	OK		278.1	1190.4	27.2	0.125	0
19	1048.8	OK		278.2	1196.1	30.8	0.11	0
20	1053.9	OK		274.8	1199.9	34.2	0.114	0
21	1053.3	OK		279	1203.5	29.1	0.115	0
22	1075.4	OK		280.4	1198.7	31.9	0.104	0
23	1065.1	OK		284.5	1218	22.9	0.094	0
24	1045	OK		281.6	1172.7	34.1	0.119	0
25	1072.2	OK		277.1	1190.4	30.2	0.112	0

Showing 25 of 60 prints. The download has all of them.

References

1. Nelson, L. S. (1984). The Shewhart control chart: tests for special causes. Journal of Quality Technology, 16(4), 237-239.
The eight special-cause tests.
2. Montgomery, D. C. Introduction to Statistical Quality Control (Wiley). Individuals and moving-range charts (sigma = average moving range / 1.128), X-bar and R charts, Cp/Cpk/Pp/Ppk.
Control-chart and capability formulas.
3. Total Materia. ASTM F2924, Ti-6Al-4V made by additive manufacturing: minimum ultimate tensile strength 895 MPa (Class C page). https://www.totalmateria.com/en-us/material/3300110
Standard minimum for comparison. Our 1000 MPa is a demonstration client requirement, not this standard.
4. Instron. ASTM A370, Mechanical Testing of Steel Products. https://www.instron.com/en/testing-solutions/astm-standards/astm-a996/
A370 covers steel; ASTM E8/E8M is the general metal tension test.
5. Chalmers University of Technology thesis on LPBF oxygen control. https://research.chalmers.se/publication/522522/file/522522_Fulltext.pdf
Oxygen level of 100 ppm or less recommended for titanium.
6. du Plessis, A. (2019). Effect of process parameters on porosity in laser powder bed fusion Ti-6Al-4V. https://blogs.sun.ac.za/duplessis/files/2020/03/effect-of-process-parameters-on-porosity-du-Plessis-2019.pdf
Plausible laser power and scan speed ranges.